Question	Answer	Marks	Partial Marks	Guidance
12(a)	$\frac{13x-4}{24}$ final answer	**3**	**M2** for $\frac{6x+9x-2x-4}{24}$ oe or **M1** for $\frac{6x+3(3x)-2(x+2)}{24}$ oe	For M2 and M1 accept equivalent fractions over a common denominator, accept as 3 separate fractions For M1 condone 1 error or allow just for numerator
12(b)	$(3x-y)(a+4y)$ final answer	**1**		Condone omission of final bracket
13	$\frac{3}{2}$ oe cao	**4**	**B3** for $16r^2 = 36$ or better or **M2** for $\pi r^2 16 = \frac{4}{3}\pi 3^3$ or **M1** for $\pi r^2 16$ or $\frac{4}{3}\pi 3^3$	Allow 3.14 etc for method
14(a)	70 **and** opposite angles of a cyclic quadrilateral sum to 180 oe	**2**	**B1** for 70 or for fully correct reason	Accept e.g. cyclic quadrilateral with no incorrect statement or angle Accept angles in opposite segments are supplementary See AG
14(b)(i)	65	**2**	**FT** 180 – 45 – *their* 70 for 2 marks **B1** for angle $BDC = 45$	May be marked on diagram Consider other angles for B1 at STM.
14(b)(ii)	65	**1**	**FT** *their* (b)(i)	
15(a)	(1, 13)	**2**	**B1** for one correct coordinate	
15(b)	$\begin{pmatrix}-9\\3\end{pmatrix}$	**2**	**B1** for $\begin{pmatrix}-9\\k\end{pmatrix}$ or $\begin{pmatrix}k\\3\end{pmatrix}$	

Question	Answer	Marks	Partial Marks	Guidance
15(c)	39	**3**	**B2** for $\lvert\overrightarrow{BA}\rvert = 13$ soi or **M2** for $3\sqrt{([-]5)^2 + ([-]12)^2}$ oe or **M1** for $([-]5)^2 + ([-]12)^2$ or $15^2 + 36^2$	
16(a)	12.5	**3**	**M2** for 51.5 – 39 OR **B1** for UQ = 51.5 **B1** for LQ = 39	Condone 1 error in a quartile for M2
16(b)	$\frac{6}{13}$ oe	**3**	**M2** for $[2\times]\left(\frac{4}{13}\times\frac{9}{12}\right)$ oe or **M1** for $\frac{4}{13}$ and $\frac{9}{12}$ **or** $\frac{9}{13}$ and $\frac{4}{12}$ If 0 scored, **SC1** for $2\left(\frac{4}{13}\times\frac{9}{13}\right)$	Accept decimal or percentage equivalents to 3sf or better. Ignore incorrect cancelling or conversion following a correct answer. Penalty –1 for 2sf or words, e.g. 1 in 2 or 1 out of 2, 0.125/1, but not ratio, i.e. not 1 to 2 SC1 implied by $\frac{72}{169}$
17	$\frac{38}{45}$ cao	**4**	**B3** for $\frac{76}{90}$ oe fraction OR **B2** for $\frac{26}{90}$ oe fraction or **M1** for 28.8…. and 2.8…. oe or for $90r = 26$ **M1** for $\frac{50}{90} + \textit{their}\ \frac{26}{90}$	**ALT method** **B3** for $\frac{76}{90}$ or **M2** for 84.4… and 8.4…. or **B1** for $0.8\dot{4}$ or 0.844…. M1 for correct conversion of $\frac{5}{9}$ and *their* $\frac{26}{90}$ to common denominator